



Project Repo

What Makes A Mobile Puzzle Game Hard?

CmpE 492 - Spring 2026 · Alperen Akyol (alperen.akyol@std.bogazici.edu.tr) · **Advisor:** Bahri Atay Özgövde



Wondering
What
Hex Fall is?
Play demo here!

The Problem Statement

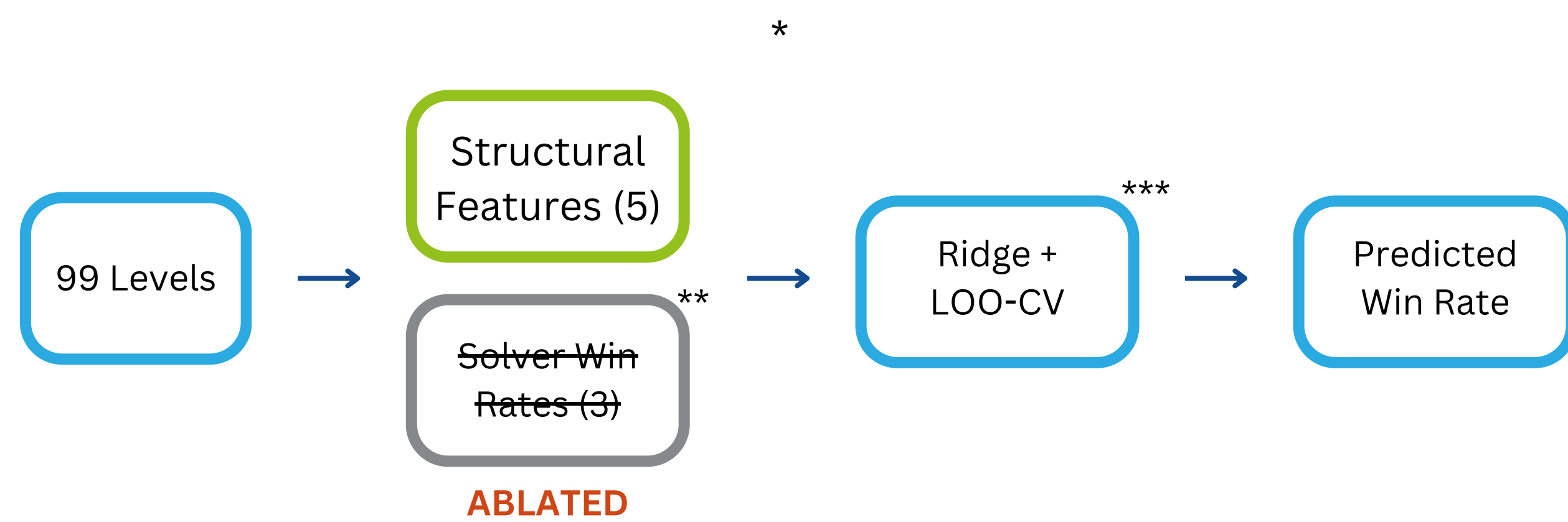
Predict a level's human win rate from the level alone — no pre-launch playtesting.

Why it matters: Procedural generation needs difficulty estimates before players see the level

Input: 99 real Hex Fall levels

Target: Real human win rate per level (from Paxie)

Bounded-Rationality Solver Ensemble



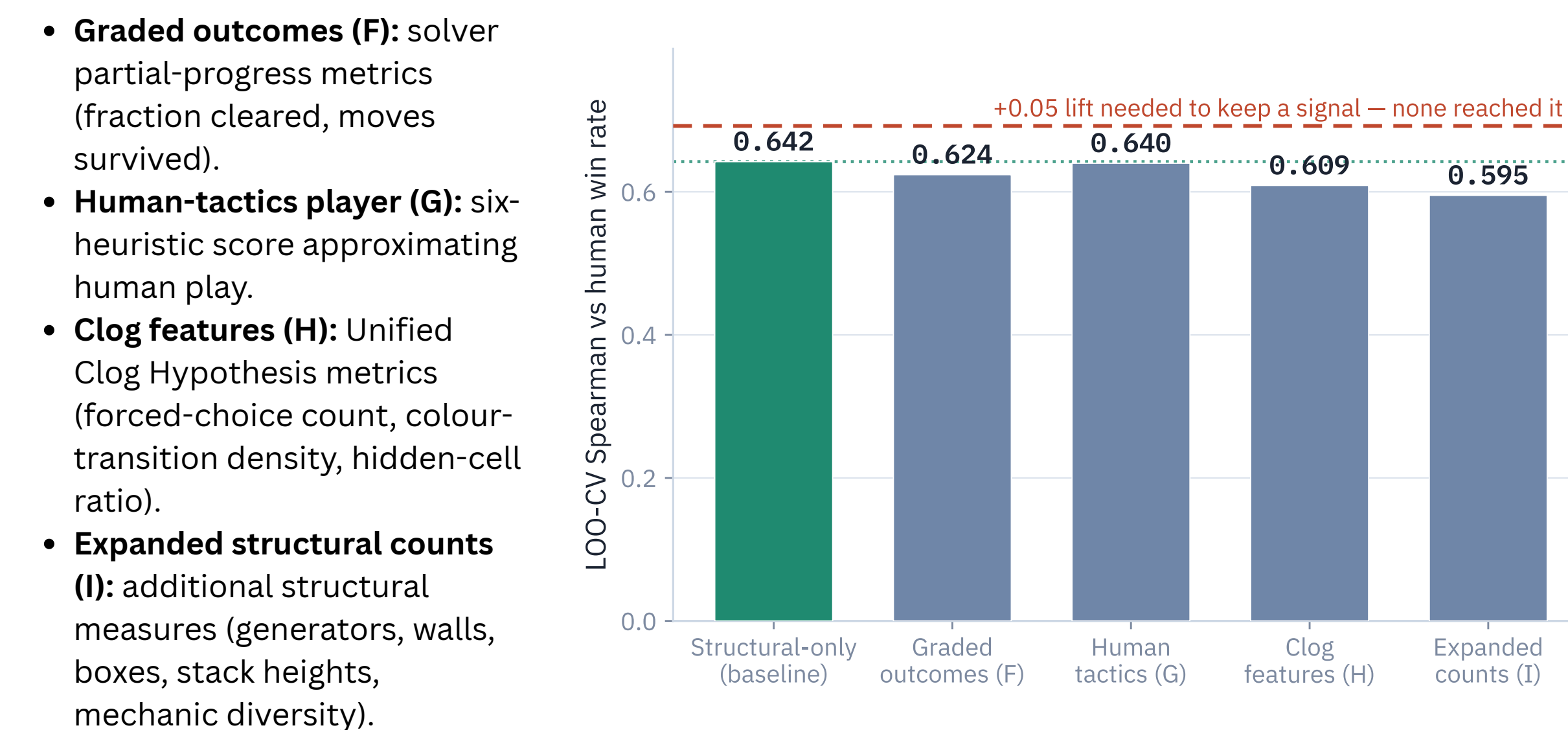
***Structural:** colours · total pieces · pins · ice · board size

****Solvers:** greedy · lookahead · MCTS

*****Fit:** Ridge regression, leave-one-out cross-validation against 99 real human winrates

Ablation Across Four Signal Axes

Four pre-registered ablations tested whether any behavioural or decision-structure signal would improve on the structural-only baseline. The acceptance threshold was a Spearman lift of +0.05.



None reached the threshold. The Clog features were verified orthogonal to level size and yet contributed no predictive signal.

Initial Methodology and Validity Objection

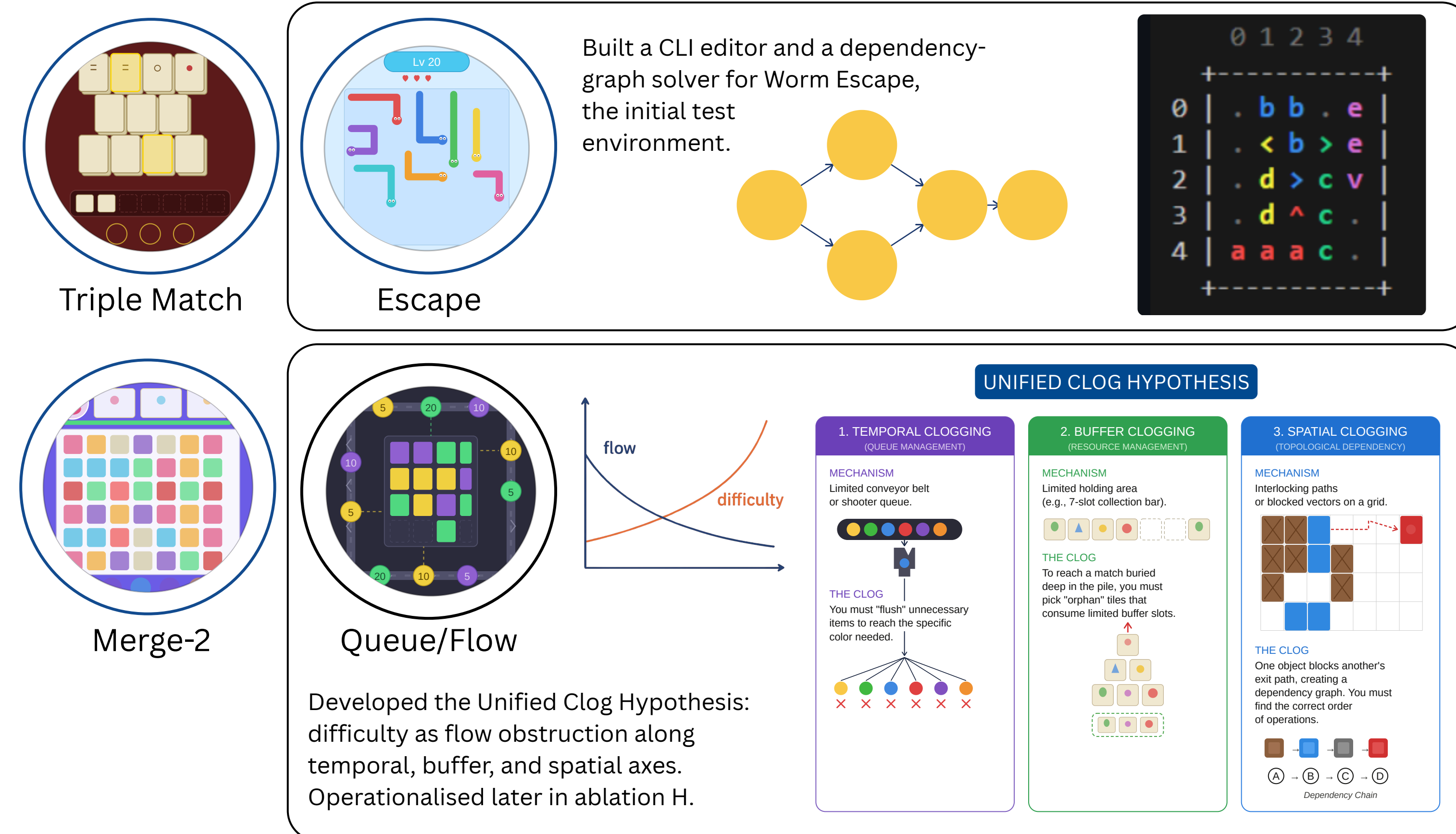
The original plan used a curiosity-driven PPO agent, with win rate as the difficulty signal. Before training began, the advisor raised three concerns:

- RL is disproportionate for a sparse-reward puzzle.
- RL agents converge toward optimal play, whereas humans do not, creating a validity mismatch between agent and human win rates.
- Reward shaping changes the objective being optimised rather than simulating different human skill levels.

Given these validity concerns, the agent-based approach was abandoned, although the simulation framework was retained.

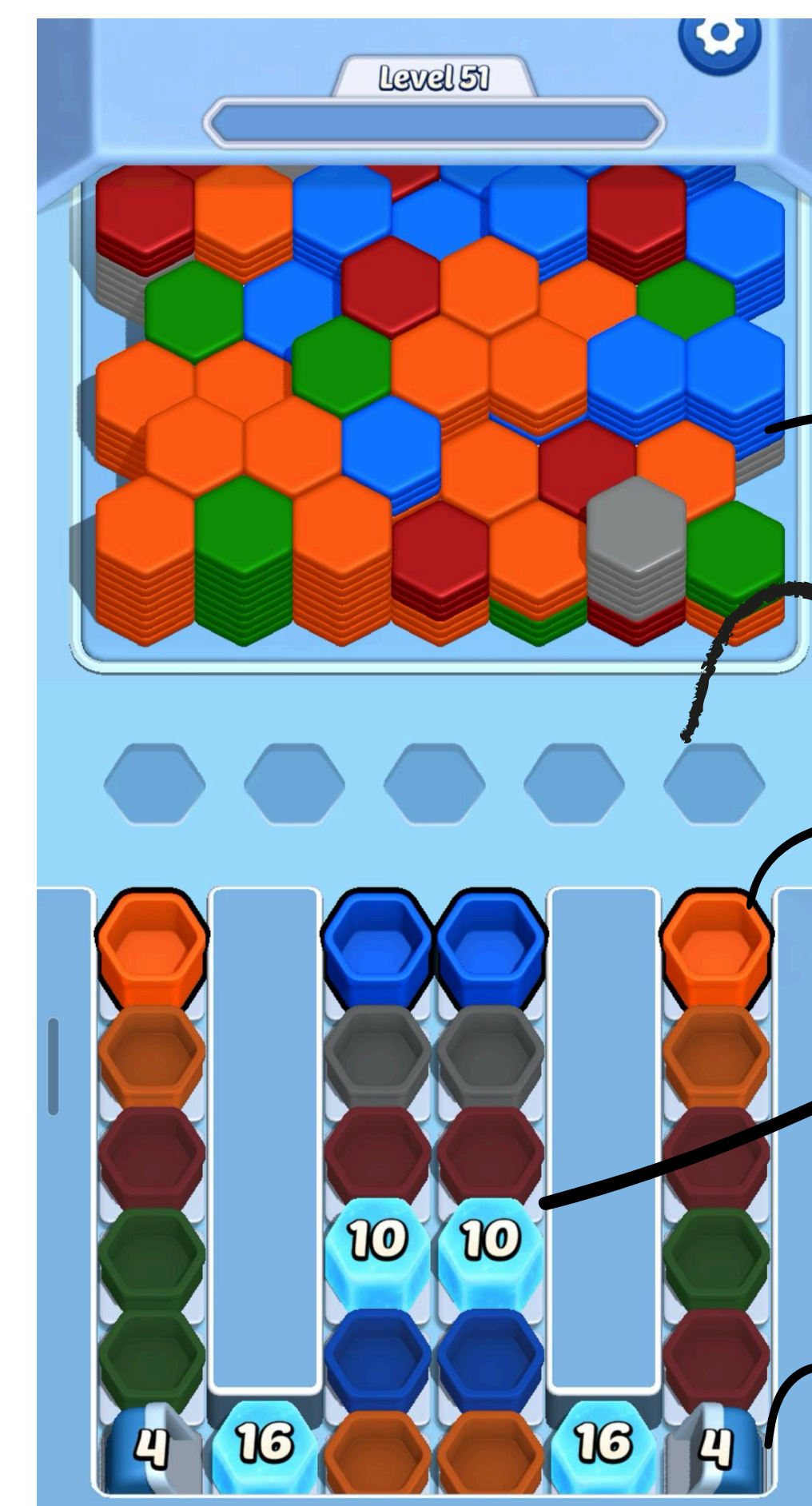
Early Phase

The project's first two months focused on a broader question: where in Paxie's portfolio is difficulty most tractable, and what makes a level hard in the first place?



The scope was narrowed to Hex Fall in early April due to higher corpus quality and the availability of per-level win-rate data provided by the Paxie CTO, which made it suitable for reliable measurement and comparative analysis.

Experimental Domain - Hex Fall



What is Hex Fall?

Hex Fall is a triple-match mobile puzzle game. The player sorts coloured slices into matching buckets while contending with several obstructions, before the board reaches saturation.

Slices: coloured pieces to be cleared.

Buffer: 5 slots for active buckets; filling all of them causes a lose state.

Bucket: Collects one colour up to 24 slices.

Ice Buckets: Frozen, thaws after a fixed number of moves.

Generator: Generates buckets in its facing direction.

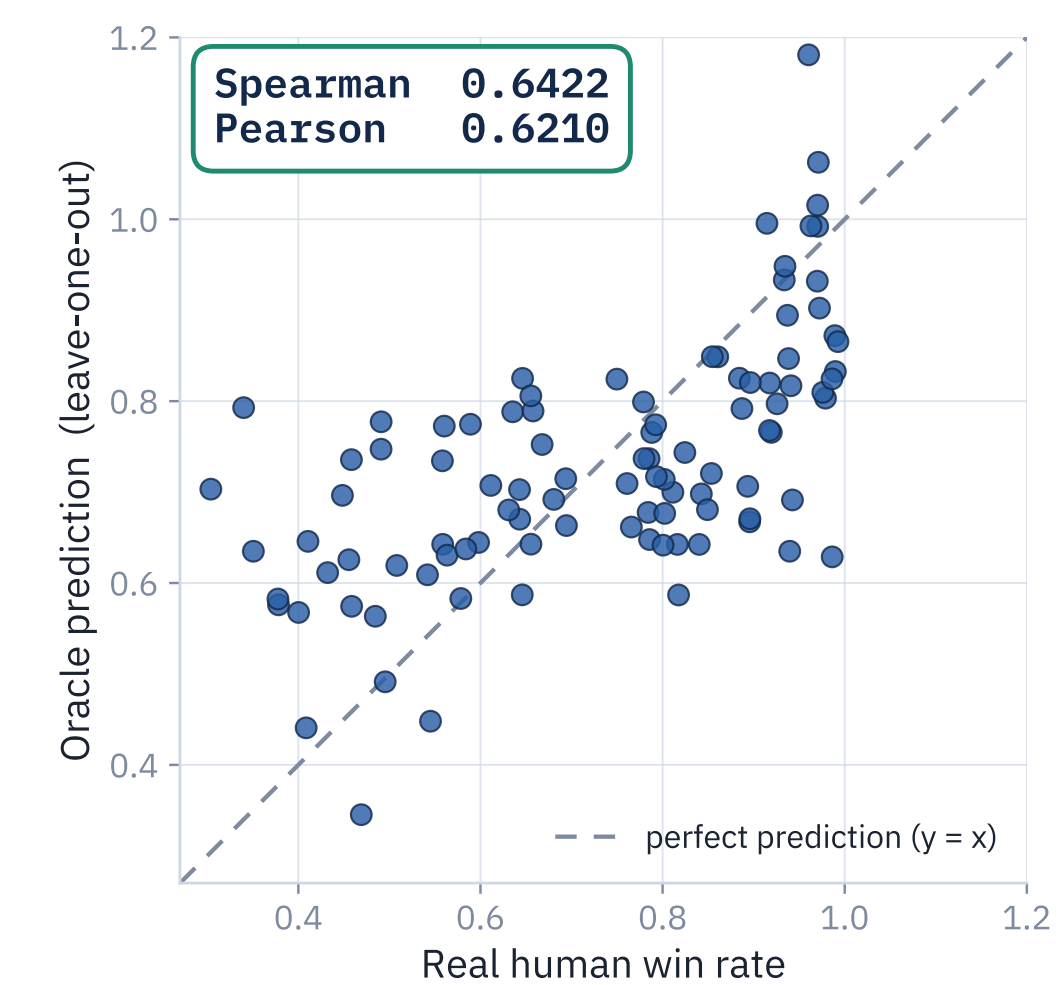
References & Acknowledgements

- Hüseyin Anil Özmen & Paxie Games (Hex Fall corpus and per-level win-rate data)
- Bergdahl, J., Gordillo, C., Tollmar, K., & Gisslén, L. (2020, August). *Augmenting automated game testing with deep reinforcement learning*.
- Ferdous, Raihana, et al. "Curiosity Driven Multi-agent Reinforcement Learning for 3D Game Testing."
- Hoerl, A. E., & Kennard, R. W. (1970). *Ridge regression: Biased estimation for nonorthogonal problems*.
- Kocsis, L., & Szepesvári, C. (2006, September). *Bandit based monte-carlo planning*.

Empirical Results

Five structural features predict human win rate on the 99-level corpus.

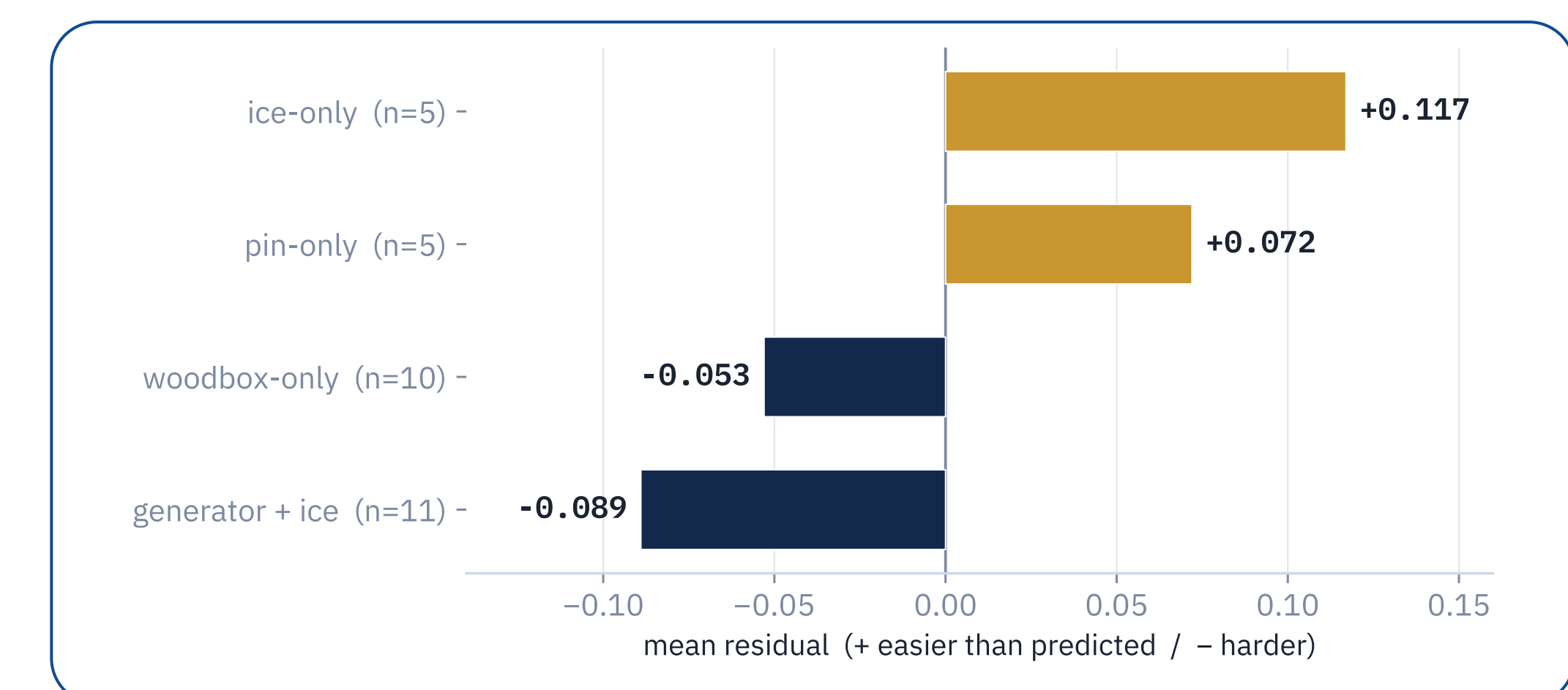
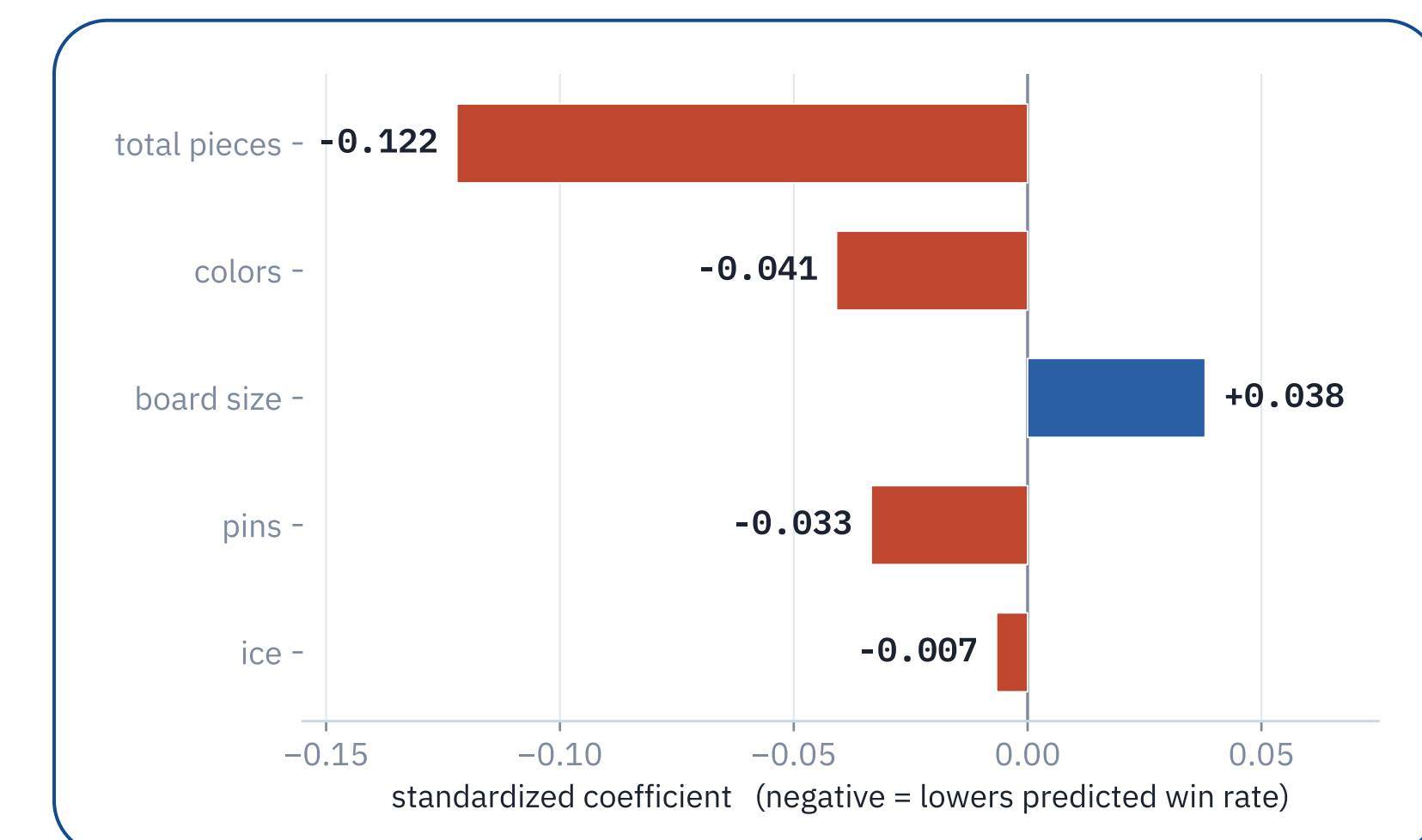
- LOO-CV Spearman 0.6422, Pearson 0.6210, on per-level win rate.
- No behavioural signal, no solver output, no tactical heuristic.
- Predicted versus real win rate, 99 levels (scatter right).



Feature Analysis

Predictive power is concentrated in a single feature, with residuals revealing mechanic-specific blind spots.

- Total slice count carries roughly three times the weight of any other coefficient (top).
- Even features verified orthogonal to slice count contribute no signal in regression.
- Difficulty on this corpus is dominated by board size.
- Residuals are not uniform across mechanics (bottom). Ice-only and pin-only levels are systematically easier than predicted; generator-plus-ice combinations are harder than the sum of parts.
- The structural features capture overall difficulty but miss specific mechanic interactions.



Limitations and Future Work

- The result is established on a single game and a single 99-level corpus. Generalisation across titles and corpora is not tested.
- Win rate saturates near 100 percent on easy levels. Average win duration is smoother and predicts at Spearman 0.96 on the same features.
- An open question is whether the structure-dominance finding holds in puzzle genres with richer decision structure, or is specific to Hex Fall.
- A second open question is whether an agent shaped toward characteristically human errors, rather than toward bounded optimality, could recover the predictive signal that bounded-rationality solvers did not.